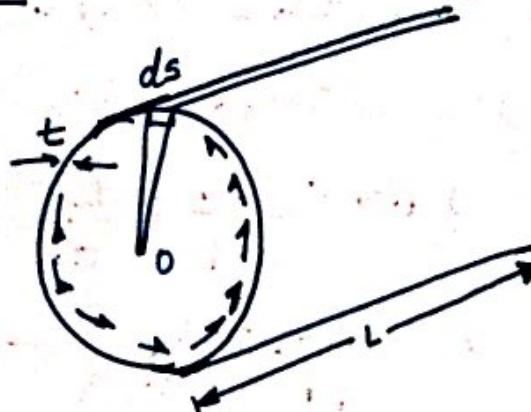


SHEAR FLOW IN CLOSED SECTIONSBREDT-BATHO THEORY

Assume a thin walled cross section shown in figure is subjected to a shear stress ' τ '.



The strain energy stored is given by,

$$U_s = \frac{\tau^2}{2G} \times \text{Volume}$$

Let us take a element ds with thickness ' t '.
Hence, the volume becomes

$$U_s = \frac{\tau^2}{2G} \times t \cdot L \cdot ds$$

The total strain energy in the tube will be,

$$U = \oint \frac{\tau^2}{2G} L t ds, \text{ now we know } q = \tau t$$

where ' q ' is the shear flow around the section

$$U = \oint \frac{\tau^2 L t^2}{2G} \frac{ds}{t}$$

$$U = \oint \frac{q^2 L}{2G t} \frac{ds}{t} = \frac{q^2 L}{2G} \oint \frac{ds}{t}$$

Force on element ds is $\sigma \times A = F$, $\tau \times (t \cdot ds)$
 $\tau \cdot t \cdot ds = q \cdot ds$

Twisting moment and θ

If the tube twists through an angle ' θ ', then the internal strain energy becomes,

$$\frac{1}{2} T \cdot \theta = \frac{q^2 L}{2G} \oint \frac{ds}{t}$$

$$\theta = \frac{q^2 L}{GT} \oint \frac{ds}{t} \quad \text{--- (1)}$$

From torsion of thin walled structures. We know,
 $\tau = \frac{T}{2tA}$, where A is area enclosed

$$\tau t = q, \quad q = \frac{T}{2A} //$$

$$\theta = \frac{T^2 L}{4A^2 TG} \oint \frac{ds}{t} = \frac{TL}{4A^2 G} \oint \frac{ds}{t}$$

From Torsion equation, $\theta = \frac{TL}{GJ}$

$$J = \frac{TL}{G\theta} = \frac{TL}{G \times \frac{TL}{4A^2 G} \oint \frac{ds}{t}} = \frac{4A^2}{\oint \frac{ds}{t}}$$

Hence, the rate of twist per unit length

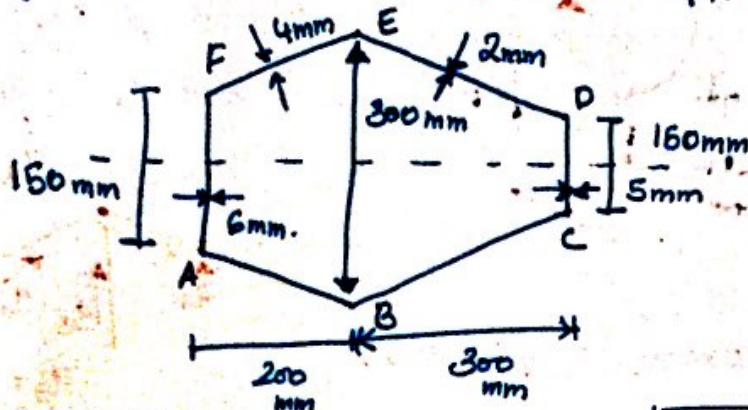
$$\frac{\theta}{L} = \frac{d\theta}{dz} = \frac{q}{2AG} \oint \frac{ds}{t} \quad [\text{From equation (1)}]$$

$$\theta = \frac{q^2 L}{GT} \oint \frac{ds}{t}, \quad q = \frac{T}{2A}, \quad T = 2qA$$

$$\theta = \frac{q^2 L}{G 2qA} \oint \frac{ds}{t} = \frac{qL}{2GA} \oint \frac{ds}{t} //$$

Example A 5m long tailplane has Uniform Cross Section Shown in fig:- Subjected to torque of 50 kN-m. (3)

Find Shear stress distribution around section and angle of twist throughout the length assuming centre of twist lies at the Centroid of the section. Take $G = 30 \text{ GPa}$.



$$AB = \sqrt{200^2 + 75^2} = 213.6 \text{ mm} \quad BC = \sqrt{300^2 + 75^2} = 309.2 \text{ mm}$$

$$\text{Area} = \frac{1}{2} (300 \times 75) \times 2 + \frac{1}{2} (200 \times 75) + (150 \times 500) = 22500 + 7500 + 75000 = 105000 \text{ mm}^2$$

$$\text{Shear stress: } \tau = \frac{q}{t}, \quad q = \frac{T}{2A} = \frac{50 \times 10^6}{2 \times 105000} = 238.09 \text{ N/mm}$$

$$\tau_{AB} = \tau_{EF} = \frac{238.09}{4} = 59.52 \text{ N/mm}^2$$

$$\tau_{ED} = \tau_{BC} = \frac{238.09}{2} = 119.04 \text{ N/mm}^2$$

$$\tau_{CD} = \frac{238.09}{5} = 47.61 \text{ N/mm}^2, \quad \tau_{AF} = 39.68 \text{ N/mm}^2$$

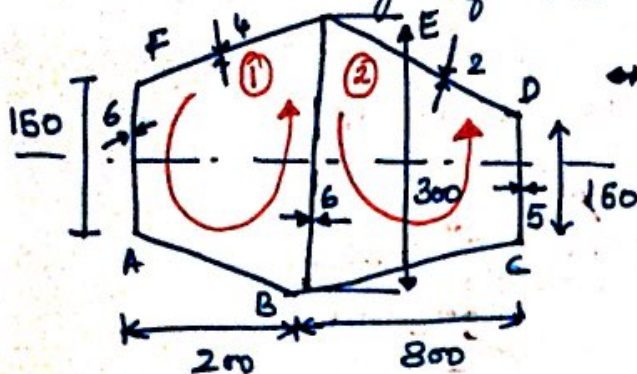
$$\text{Angle of twist} = \theta = \frac{TL}{GJ}, \quad \text{where } J = \frac{4A^2}{\oint \frac{ds}{t}}$$

$$\oint \frac{ds}{t} = \left[\frac{150}{6} \right] + 2 \left[\frac{213.6}{4} \right] + 2 \left[\frac{309.2}{2} \right] + \left[\frac{150}{5} \right] = 417.6$$

$$J = \frac{4A^2}{417.6} = \frac{4(105000)^2}{417.6} = 10.5 \times 10^7 \text{ mm}^4, \quad \text{Substituting } \theta = 0.078 \text{ rad or } 4.44^\circ //$$

Example:-

The tail-plane is modified to form a 2-cell section as shown. Find the shear stress distribution and the angle of twist over a 5m length of section. $G = 30 \text{ GPa}$



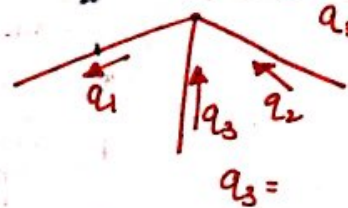
Shear flow in Section

① and ② be,

q_1 and q_2 respectively

$$q_2 + q_3 = q_1$$

$$q_3 = q_1 - q_2$$



$$A_1 = \left(\frac{1}{2} \times 200 \times 75 \right) 2 + (150 \times 200) = 45,000 \text{ mm}^2$$

$$A_2 = \left(\frac{1}{2} \times 800 \times 75 \right) 2 + (150 \times 800) = 67,500 \text{ mm}^2$$

To find $\oint \frac{ds}{t}$ to substitute in $\frac{d\theta}{dz} = \frac{q}{2AG} \oint \frac{ds}{t}$, we construct a table.

WALL	L	t	ds/t	q	$\frac{ds}{t} \cdot q$ (N/mm)
AB	200	6	33.3	q_1	$33.3 q_1$
BE	300	4	75	$q_1 - q_2$	$75(q_1 - q_2)$
AF	150	4	37.5	q_1	$37.5 q_1$
DE	800	4	200	q_1	$200 q_1$

Table for Cell 1

$$\Sigma = 181.8 q_1 - 75 q_2 = q \oint \frac{ds}{t}$$

Hence, $\frac{d\theta}{dz} = \frac{q}{2AG} \oint \frac{ds}{t}$

$$\frac{d\theta}{dz} = \frac{181.8 q_1 - 75 q_2}{2(45,000) \times 30 \times 10^9}$$

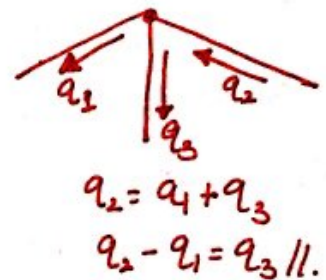
Equation ①

Solving for Section (2).

(5)

Wall	L	t	ds/t	q	q.ds/t (N/mm)
BC	309.2	2	154.6	q_2	$154.6 q_2$
CD	150	5	30	q_2	$30 q_2$
DE	309.2	2	154.6	q_2	$154.6 q_2$
EB	300	6	50	$q_2 - q_1$	$50(q_2 - q_1)$
					$\Sigma = -50q_1 + 389.2q_2$

Hence $\frac{dq}{dz} = \frac{389.2q_2 - 50q_1}{2 \times 67500 \times G}$



For compatibility:- $\left[\frac{dq}{dz}\right]_1 = \left[\frac{dq}{dz}\right]_2$

$$\frac{181.8q_1 - 50q_2}{90000 G} = \frac{389.2q_2 - 50q_1}{135000 G}$$

$$272.7q_1 - 75q_2 = 389.2q_2 - 50q_1$$

Solving $q_1 = 1.44q_2$ — (1)

Since,

$$T = 2Aq, \quad 50 \times 10^6 = 2(45000q_1 + 67500q_2) \quad (2)$$

Solving equation (1) and (2), we get.

$$q_2 = 189 \text{ N/mm}$$

$$q_1 = 272 \text{ N/mm}$$

(6)

To find Angle of twist.

$$\left[\frac{d\theta}{dz} \right]_1 = \frac{181.8 q_1 - 50 q_2}{90000 G}$$

Solving by Substituting q_1 and q_2

$$\left[\frac{d\theta}{dz} \right]_1 = 0.849^\circ / m. \quad \text{for } 5m \Rightarrow \theta = 4.24^\circ.$$

$$\left[\frac{d\theta}{dz} \right]_2 = \frac{389.2 q_2 - 50 q_1}{1,35,000 G}$$

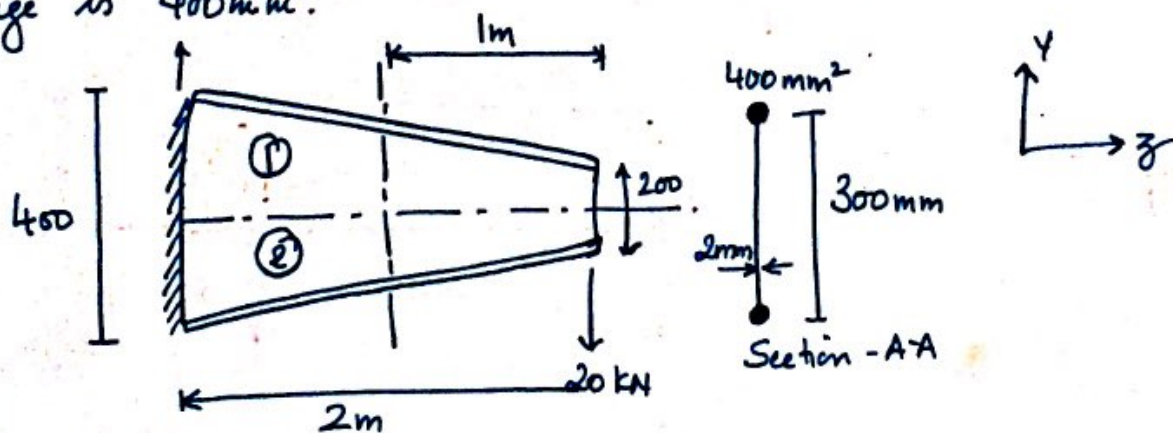
Problems on Tapers-Spanwise effect:-

(7)

Most of the times the cross section of the beam is constant. To reduce weight in aerospace structures the beams are tapered to achieve maximum structural efficiency. The variation in cross section causes large errors in shear stresses.

Problems:-

Determine shear flow distribution in the web of the tapered beam at Section A-A along the mid of its length. The web of beam has thickness 2mm and fully effective in resisting direct stresses. The beam tapers symmetrically about its horizontal axis. The cross sectional area of each flange is 400 mm^2 .



Bending Moment due to externally applied load.

$$M_x = 20 \times 1 \text{ m} = 20 \text{ kN-m.} \quad \& \quad S_y = -20 \text{ kN-m}$$

$$M_y = 0 \quad I_{xy} = 0.$$

$$\text{Hence, } \sigma_x = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y.$$

$$\sigma_z = \frac{M_x I_{yy} \cdot y}{I_{xx} I_{yy}} = \frac{M_x \cdot y}{I_{xx}}$$

Calculating Moment of Inertia about x axis

$$I_{xx} = \frac{\pi d^4}{64} + Ay^2 + \frac{bt^3}{12} = 22.5 \times 10^6 \text{ mm}^4$$

$$\sigma_{z1} = \frac{20 \times 10^6 \times 150}{22.5 \times 10^6} = 133.3 \text{ N/mm}^2 = -\sigma_{z2}$$

Axial loads $P_{z1} = -P_{z2} = 133.3 \times 400 = 53320 \text{ N}$
 $\sigma \times A$

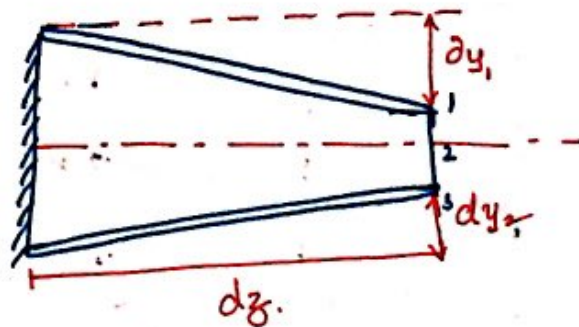
Shear load resisted by the beam is,
 Given by.

$$S_{y,w} = S_{y1} + P_{z1} \frac{dy_1}{dz} + P_{z2} \frac{dy_2}{dz}$$

$$S_{y,w} = -20 \times 10^3 - 53320 \frac{dy_1}{dz} + 53320 \frac{dy_2}{dz}$$

$$\frac{dy_1}{dz} = \frac{-100}{2 \times 10^3} = -0.05$$

$$\frac{dy_2}{dz} = \frac{100}{2 \times 10^3} = 0.05$$



$$S_{y,w} = -14668 \text{ N (after Substitution)}$$

$$Q_s = \frac{-S_{y,w}}{I_{xx}} \left[\int_0^s t \cdot y \cdot ds + B_1 Y_1 \right] \text{ or } \frac{-S_{y,w}}{I_{xx}} \left(\int_0^s t \cdot y \cdot ds + B_2 Y_2 \right)$$

$$B = 400 \quad t = 2$$

$$Y = 150 \quad y = (150 - s)$$

Substitute the values, integrate and solve for Q_s (from 1-2)

Airy's STRESS FUNCTION:

(9)

A Solving 3 equations for 3 unknown's can be reduced to a simplex form. 3 equations can be reduced to a single equation containing a function $\phi(x, y)$ which is called the airy's stress function.

The airy's stress function will satisfy the equilibrium equations when the following are true.

$$\sigma_{xx} = \frac{\partial^2 \phi}{\partial y^2}, \quad \sigma_{yy} = + \frac{\partial^2 \phi}{\partial x^2}, \quad \sigma_{xy} = - \frac{\partial^2 \phi}{\partial x \partial y}. \quad \text{--- (1)}$$

The equilibrium equations are: (which are satisfied by the above equation)

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0, \quad \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} = 0. \quad \text{--- (2)}$$

to satisfy the compatibility conditions

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (\sigma_x + \sigma_y) = 0 \quad \text{--- (3)}$$

$$\frac{\partial(\partial^2 \phi)}{\partial x(\partial y)^2} + \frac{\partial(-\partial^2 \phi)}{\partial x(\partial y)(\partial y)} = 0$$

$$\frac{\partial^3 \phi}{\partial x \partial y^2} - \frac{\partial^3 \phi}{\partial x \partial y^2} = 0$$

Substituting eqn (1) in (2), we ~~get~~ can see that it satisfies the equilibrium equations.

Further substitution in (3), we get,

$$\frac{\partial^4 \phi}{\partial x^4} + \frac{2 \partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} = 0$$

Example :-

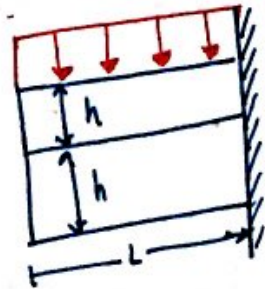
(10)

A Cantilever of length L and depth $2H$ is in a state of plane stress. The cantilever is of unit thickness and supported rigidly at $x=L$ and loaded as shown in fig. Prove that the stress function

$$\phi = Ax^2 + Bx^2y + Cy^3 + D(5x^2y^3 - y^5)$$

is valid for the beam and Evaluate Constants A, B, C & D .

Sol:- Stress function must satisfy



$$\textcircled{1} - \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} = 0.$$

$$\frac{\partial \phi}{\partial x} = 2Ax + 2Bxy + 10xy^3D = \sigma_x$$

$$\frac{\partial^2 \phi}{\partial x^2} = 2A + 2By + 10y^3D = \sigma_y, \quad \frac{\partial^3 \phi}{\partial x^3} = 0, \quad \frac{\partial^4 \phi}{\partial x^4} = 0,$$

$$\text{And, } \frac{\partial \phi}{\partial y} = Bx^2 + 3Cy^2 + 15x^2y^2D - 5Dy^4, \quad \frac{\partial^3 \phi}{\partial y^3} = 6C + 30x^2D - 60Dy^2$$

$$\frac{\partial^2 \phi}{\partial y^2} = 6Cy + 30x^2yD - 20Dy^3 = \tau_{xy}, \quad \frac{\partial^4 \phi}{\partial y^4} = -120Dy$$

$$\frac{\partial^2 \phi}{\partial y^2}$$

$$\frac{\partial^2 \phi}{\partial x \partial y} = 2Bx + 30Dxy, \quad -\tau_{xy} = \frac{\partial^2 \phi}{\partial x \partial y}$$

$$\frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 60Dy, \quad \text{Substituting in equation } \textcircled{1}.$$

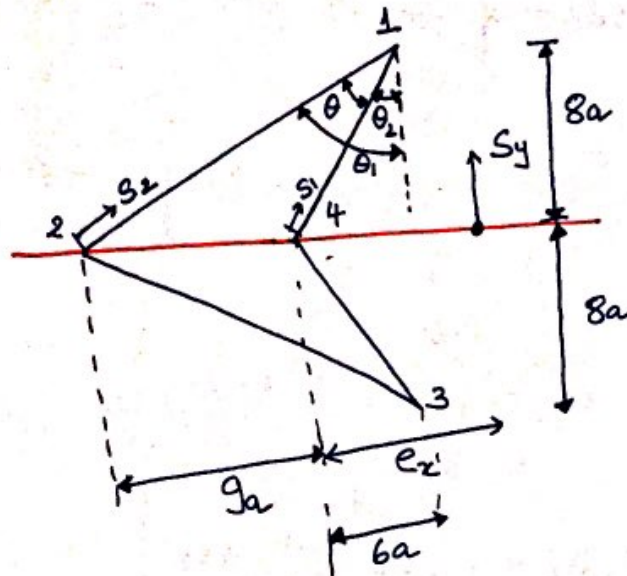
$$\text{where, } \frac{\partial^4 \phi}{\partial x^4} = 0, \quad \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 60Dy, \quad \frac{\partial^4 \phi}{\partial y^4} = -120Dy$$

Applying Boundary Conditions, the problem can be solved.

Shear Centre of closed Sections:

(11)

A thin walled closed section beam is shown in fig. Each wall has thickness t and shear modulus G . Calculate the distance of shear centre from point 4.



~~Sol~~ Simplification of the problem. $I_{xy} = 0$, $S_x = 0$, hence

$$Q_s = -\frac{S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds + Q_{sp}$$

length of 2-3 = $\sqrt{15^2 + 8^2} = 17$, length of 4-3 = $\sqrt{6^2 + 8^2} = 10$

Similarly on the other section as the cross section is symmetric.

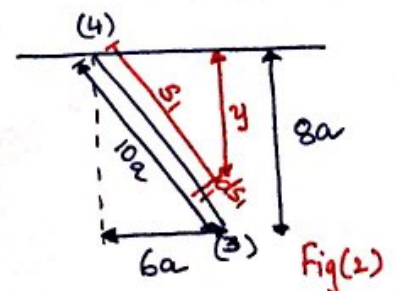
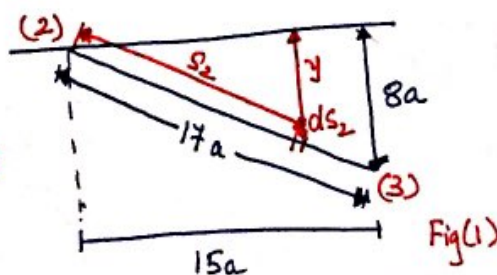
$I_{xx} = \int y^2 \cdot dA = \int y^2 \cdot t \cdot ds$, Now for 2-3 and 3-4, we

have the elemental length ds_2 and ds_1 respectively.

I_{xx} for 2-3 is $= \int_0^{17a} y^2 \cdot t \cdot ds_2$ and for 4-3 $\int_0^{10a} y^2 \cdot t \cdot ds_1$

Figure 2 indicates
Section 4-3

and figure 1 indicates
Section 2-3



From Similarity of Triangles, we can get 'y' to be

(12)

$$\frac{8x}{17a} = \frac{y}{s_2}, \quad y = \frac{8}{17} \cdot s_2 \quad \text{and similarity for section 4-3}$$

$y = \frac{8}{10} s_1$. Substituting in, to calculate I_{xx} .

$$I_{xx} = 2 \left[\int_0^{10a} \left(\frac{8}{10} s_1 \right)^2 t \cdot ds_1 + \int_0^{17a} \left(\frac{8}{17} s_2 \right)^2 t \cdot ds_2 \right], \text{ evaluating this term.}$$

$$I_{xx} = 2 \left[\int_0^{10a} 0.64 s_1^2 t \cdot ds_1 + \int_0^{17a} 0.22 s_2^2 t \cdot ds_2 \right]$$

$$I_{xx} = 2 \left[0.64 t \int_0^{10a} s_1^2 \cdot ds_1 + 0.22 t \int_0^{17a} s_2^2 \cdot ds_2 \right]$$

$$I_{xx} = 2 \left[0.64 t \left[\frac{s_1^3}{3} \right]_0^{10a} + 0.22 t \left[\frac{s_2^3}{3} \right]_0^{17a} \right] = 2 \left[0.64 t \left[\frac{(10a)^3}{3} \right] + 0.22 t \left[\frac{(17a)^3}{3} \right] \right]$$

Simplifying, we get,

$$I_{xx} = 2 \left[213.33 a^3 t + 360.28 a^3 t \right] = \boxed{1147 a^3 t}$$

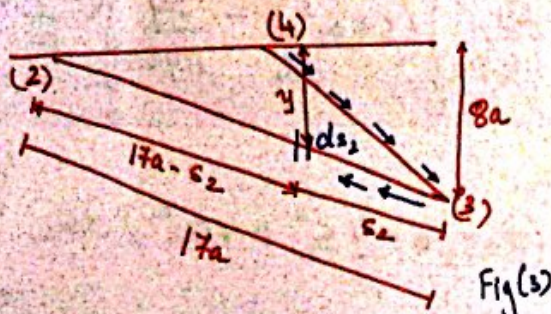
Calculating shear flow in 4-3 and 2-3,

$$q_{4-3} = -\frac{S_y}{I_{xx}} \int_0^{s_1} \left(\frac{8}{10} ds_1 \right) \cdot t \cdot ds = -\frac{S_y (0.8)}{I_{xx}} \int_0^{s_1} t \cdot s_1 \cdot ds = -\frac{S_y \cdot t \cdot (0.8) s_1^2}{2 \cdot I_{xx}}$$

$$q_{4-3} = \frac{-S_y \cdot t \cdot (0.4) \cdot s_1^2}{1147 a^3 t} = \frac{-S_y (0.4) s_1^2}{1147 \cdot a^3} //$$

Similarly in 2-3,

$$q_{23} = \frac{-S_y}{1147a^3t} \int_0^{s_2} t \cdot \left(\frac{8}{17} (17a - s_2) \right) \cdot ds_2 + q_{34}$$



From fig(3), $\frac{y}{8a} = \frac{(17a - s_2)}{17a}$

$$y = \frac{8}{17} (17a - s_2)$$

Substituting for q_{34} , $q_{34} = \frac{-S_y (0.4) (s_1^2)}{1147a^3}$ (As derived earlier)
 $s_1 = 10a$

$$q_{34} = \frac{-S_y (0.4) (10a)^2}{1147a^3}$$

$$q_{34} = \frac{-S_y (0.4) (100a^2)}{1147a^3} = \frac{-S_y 40}{1147a} \quad \text{or} \quad \frac{-S_y}{1147a^3} [40a^2]$$

Substituting back, we get,

$$q_{23} = \frac{-S_y t}{1147a^3 t} \int_0^{s_2} \frac{8}{17} (17a - s_2) ds_2 + \frac{-S_y}{1147a^3} [40a^2]$$

$$q_{23} = \frac{-S_y}{1147a^3} \int_0^{s_2} \left(8a - \frac{8}{17} s_2 \right) ds_2 + \frac{-S_y}{1147a^3} [40a^2]$$

$$q_{23} = \frac{-S_y}{1147a^3} \left[8as_2 - \frac{8}{17} \frac{s_2^2}{2} + 40a^2 \right]$$

Shear flow in segment 2-3 is given by, the above equation.

The total shear flow q_{so} is given by,

(14)

$$q_{so} = - \frac{\oint q_s \cdot ds}{\oint ds}$$

$$So, \oint ds = (17a + 10a) \times 2 = 54a$$

$$q_{so} = - \frac{\oint q_s \cdot ds}{\oint ds} = \frac{-1}{54a} \left[\frac{-S_y}{1147a^3} \int_0^{10a} (0.4s_1^2) ds_1 - \frac{S_y}{1147a^3} \int_0^{17a} \left(\frac{-4}{17} s_2^2 + 8as_2 + 40a^2 \right) ds_2 \right]$$

for symmetry.

$$q_{so} = \frac{2 \cdot S_y}{54a \times 1147a^3} \left[\int_0^{10a} 0.4s_1^2 ds_1 + \int_0^{17a} \left(8as_2 + \frac{-4}{17} s_2^2 + 40a^2 \right) ds_2 \right]$$

Integrating :-

$$q_{so} = \frac{2 \cdot S_y}{54a \times 1147a^3} \left[0.4 \frac{(10a)^3}{3} + 4a(17a)^2 - \frac{4}{17} \frac{(17a)^3}{3} + 40a^2(17a) \right]$$

$$q_{so} = \frac{2 \cdot S_y}{54 \times 1147a^4} \left[133.33a^3 + \frac{1156}{3}a^3 - 385.33a^3 + 680a^3 \right]$$

$$q_{so} = \frac{S_y a^3}{30,969a^4} \left[\frac{1584}{19.55} \right] = \frac{S_y}{19.55a} = \frac{S_y}{1147a^3} (58.7a^2) = q_{so}$$

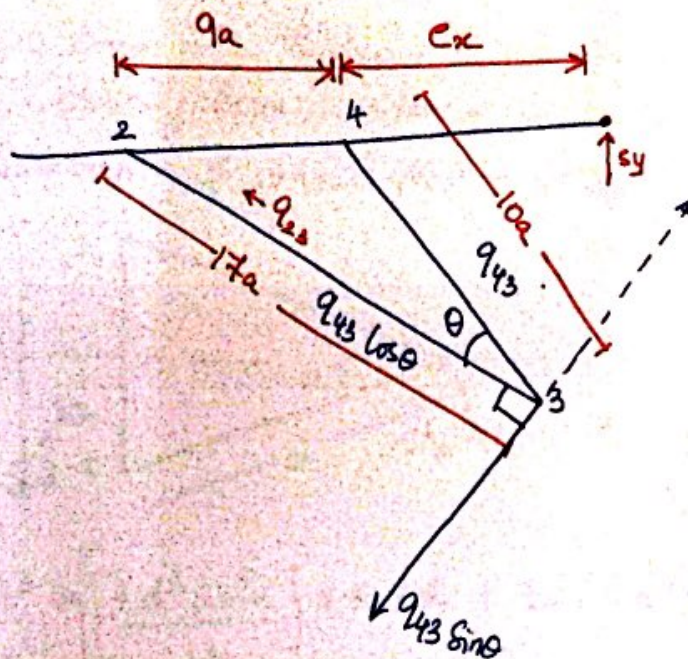
Taking  Total $q_{43} = - \frac{S_y(0.4)s_1^2}{1147a^3} + \frac{S_y}{19.55a}$

$$\text{and Total } q_{23} = - \frac{S_y \left[8as_2 - \frac{4}{17} s_2^2 + 40a^2 \right]}{1147a^3} + \frac{S_y}{19.55a}$$

Taking moments about point 2 to avoid equation q_{2-3} which is complex compared to q_{43} .

Taking moments about point 2:

15



$$S_y(e_x + 9a) = 2 \int_0^{10a} q_{43} \sin \theta \cdot 17a \cdot ds_1$$

$$S_y(e_x + 9a) = -\frac{2 S_y}{1147 a^3} \cdot 17a \sin \theta \int_0^{10a} (0.4 s_1^2 + 58.7 a^2) ds_1$$

$$S_y(e_x + 9a) = \frac{-34 S_y \cdot a \sin \theta}{1147 a^3} \int_0^{10a} (0.4 s_1^2 + 58.7 a^2) ds_1$$

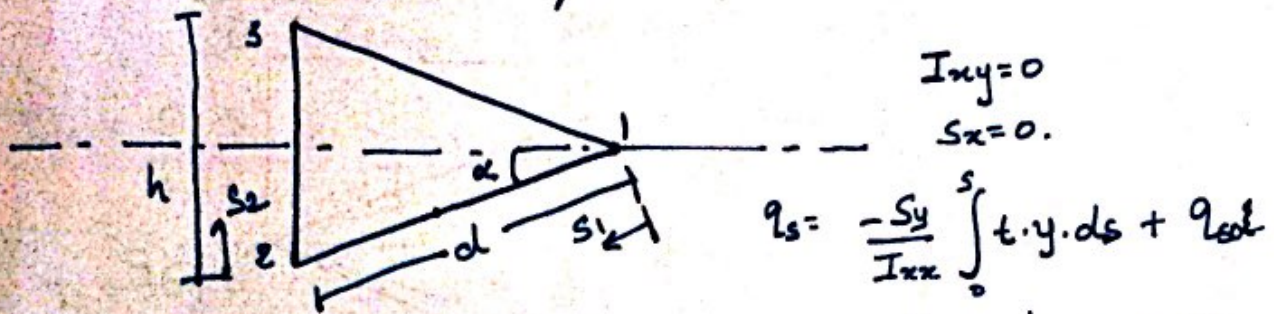
$$S_y(e_x + 9a) = \frac{-34 a \sin \theta \cdot S_y}{1147 a^3} \left[\frac{0.4 s_1^3}{3} + 58.7 a^2 [10a] \right]_0^{10a}$$

$$S_y(e_x + 9a) = \frac{-34 a \sin \theta S_y}{1147 a^3} \left[\frac{0.4 [10a]^3}{3} + 58.7 a^2 [10a] \right]$$

$$e_x + 9a = \frac{-34 a^4 \sin \theta}{1147 a^3} [720] = 21.35 a \sin \theta$$

$$e_x = 21.35 a \sin \theta - 9a //$$

A thin walled beam with thickness t in the shape of an isosceles triangle is shown below, with a load S_y at the apex. Calculate distribution of Shear flow over the cross section.



$$I_{xy} = 0$$

$$S_x = 0.$$

$$q_s = -\frac{S_y}{I_{xx}} \int_0^s t \cdot y \cdot ds + q_s^0$$

$$I_{xx} = \frac{th^3}{12} + \left[\frac{td^3 \sin^2 \alpha}{12} + td \left(\sin^2 \alpha \frac{d^2}{4} \right) \right] 2 \quad \left| \begin{array}{l} \sin \alpha = \frac{y}{d/2} \\ y = \sin \alpha \frac{d}{2} \end{array} \right.$$

$$I_{xx} = \frac{th^3}{12} + 2 \left[\frac{td^3 \sin^2 \alpha}{12} + \frac{td^3 \sin^2 \alpha}{4} \right] =$$

$$I_{xx} = \frac{th^3}{12} + \left[\frac{td^3 \sin^2 \alpha}{6} + \frac{td^3 \sin^2 \alpha}{2} \right] = \frac{th^3}{12} + \left[\frac{td^3 h^2}{6 \cdot 4 d^2} + \frac{td^3 h^2}{2 \cdot 4 d^2} \right]$$

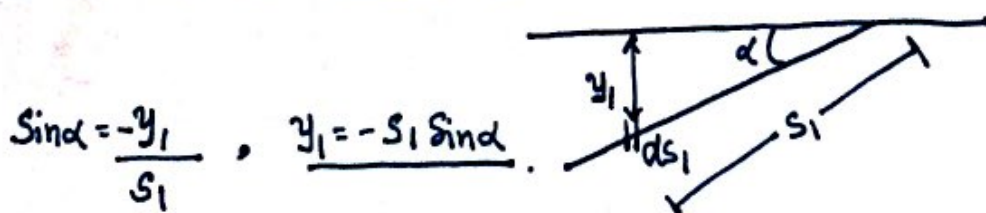
$$I_{xx} = \frac{th^3}{12} + \left[\frac{th^2 d}{24} + \frac{th^2 d}{8} \right] = \frac{th^3}{12} + \left[\frac{8th^2 d + 24th^2 d}{192} \right]$$

$$I_{xx} = \left[\frac{th^3}{12} + \frac{32th^2 d}{192} \right], \quad I_{xx} = \left[\frac{th^3}{12} + \frac{th^2 d}{6} \right], \quad \text{Also } \frac{th^3}{12} + \frac{th^2 d}{6}$$

$$I_{xx} = th^2 \left[\frac{h}{12} + \frac{d}{6} \right], \quad I_{xx} = th^2 \left[\frac{6h + 12d}{72} \right], \quad I_{xx} = th^2 \left[\frac{6h + 12d}{72} \right]$$

$$I_{xx} = \frac{th^2}{72} [6h + 12d], \quad I_{xx} = \frac{th^2}{12} [6h + 2d] //$$

Let us cut at Section 1.



$$\sin \alpha = \frac{-y_1}{s_1}, \quad y_1 = -s_1 \sin \alpha$$

$$Q_{12} = \frac{-S_y}{I_{xx}} \int_0^{s_1} t \cdot (-s_1 \sin \alpha) ds_1 = \frac{+S_y t \cdot \sin \alpha}{I_{xx}} \int_0^{s_1} s_1 ds_1$$

$$Q_{12} = \frac{S_y \cdot t \cdot \sin \alpha \cdot s_1^2}{2 I_{xx}} = \frac{S_y \cdot \cancel{t} \cdot \cancel{h} \cdot s_1^2}{\cancel{4d} \cdot \frac{\cancel{h}^2}{12} [h+2d]}$$

$$Q_{12} = \frac{S_y \cdot s_1^2 \cdot 3}{hd[h+2d]}, \text{ at point (2), } s_1 = d$$

$$Q_2 = \frac{3S_y \cdot d^2}{hd[h+2d]} = \frac{3 \cdot S_y \cdot d}{h[h+2d]}$$

$$Q_{23} = Q_{23} + Q_2, = \frac{-S_y}{I_{xx}} \int_0^{s_2} t \cdot y \cdot ds + Q_2$$

$$Q_{23} = \frac{-S_y}{I_{xx}} \int_0^{s_2} t \cdot \left(-\frac{h}{2} + s_2\right) ds_2 + \frac{3S_y \cdot d}{h[h+2d]}$$

$$Q_{23} = \frac{-S_y \cdot t}{I_{xx}} \left[\int_0^{s_2} -\frac{h}{2} ds_2 + \int_0^{s_2} s_2 ds_2 \right] + \frac{3S_y \cdot d}{h[h+2d]}$$

$$Q_{23} = \frac{-S_y \cdot t}{I_{xx}} \left[-\frac{hs_2}{2} + \frac{s_2^2}{2} \right] + \frac{3 \cdot S_y \cdot d}{h[h+2d]}$$

$$Q_{23} = \frac{-S_y \cdot \cancel{t} \cdot s_2}{\cancel{2} \cdot \frac{\cancel{h}^2}{12} [h+2d]} \left[\frac{-h + s_2}{2} \right] + \frac{3 \cdot S_y \cdot d}{h[h+2d]}$$

$$Q_{23} = \frac{S_y \cdot s_2 \cdot 6}{h[h+2d]} \left[\frac{1 - s_2/h}{2} \right] + \frac{3S_y \cdot d}{h[h+2d]}$$

$$Q_{23} = \frac{S_y \cdot 3}{h[h+2d]} \left[2s_2 - \frac{2s_2^2}{h} + d \right]$$

If the moment center coincides with line of action of S_x & S_y then,

$0 = \oint P \cdot Q \cdot ds + 2A q_{so}$, where P is the $\perp^{\text{th}}$ distance from moment point.

Taking moments about point ①.

$$0 = \int_0^h d \cos \alpha \cdot q_{23} \cdot ds + 2 \cdot \frac{1}{2} h \times d \cos \alpha \cdot q_{so}$$

$$0 = \int_0^h q_{23} ds + h \cdot q_{so}, \text{ Substituting } q_{23} \text{ from derived equation.}$$

$$0 = \int_0^h \frac{3S_y}{h[h+2d]} \left[2S_2 - \frac{2S_2^2}{h} + d \right] ds + h \cdot q_{so}$$

$$0 = \frac{3S_y}{h[h+2d]} \left[\frac{2S_2^2}{2} - \frac{2S_2^3}{3h} + d \cdot S_2 \right]_0^h + h \cdot q_{so}$$

$$\text{or } h \cdot q_{so} = \frac{3 \cdot S_y}{h[h+2d]} \left[\frac{2S_2^3}{3h} - S_2^2 - d \cdot S_2 \right]_0^h$$

$$h \cdot q_{so} = \frac{3S_y}{h[h+2d]} \left[\frac{2h^3}{3h} - h^2 - d \cdot h \right] = \frac{3 \cdot S_y h}{h[h+2d]} \left[\frac{2h}{3} - h - d \right]$$

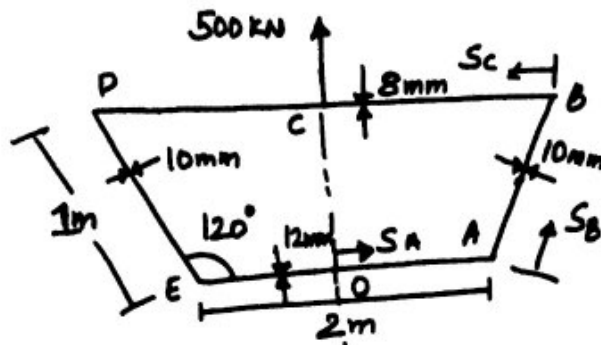
$$q_{so} = \frac{3 \cdot S_y}{[h+2d]} \left[\frac{2}{3} - 1 - \frac{d}{h} \right] = \frac{3 \cdot S_y}{[h+2d]} \left[-\frac{1}{3} - \frac{d}{h} \right] \neq \frac{3 \cdot S_y}{h[h+2d]} \left[\frac{h-d}{3} \right]$$

$$q_{so} = \frac{3 \cdot S_y}{[h+2d]} \left[\frac{2-3}{3} - \frac{d}{h} \right] = \frac{3 \cdot S_y}{[h+2d]} \left[-\frac{1}{3} - \frac{d}{h} \right] = \frac{3 \cdot S_y}{[h+2d]} \left[\frac{-h-3d}{3h} \right]$$

$$q_{so} = \frac{-S_y [h+3d]}{h[h+2d]} //$$

A box girder has a cross section shown in fig. (19)

It shows a vertical shear load of 500 kN applied through its shear centre. Calculate the shear flow distribution and the maximum shear stress in the section.



$$DB = 3\text{m}$$

$$\text{Area of } DB = 3 \times 10^3 \times 8 = 24 \times 10^3 \text{ mm}^2$$

$$EA = 2\text{m}$$

$$\text{Area of } EA = 2 \times 10^3 \times \frac{12}{2} = 24 \times 10^3 \text{ mm}^2$$

$$\text{Area of } EA = \text{Area of } BA = 1 \times 10^3 \times 10 = 10 \times 10^3 \text{ mm}^2$$

And since, $S_y = 500 \text{ kN}$, $S_x = 0$ and $I_{xy} = 0$,

$$q_s = - \frac{500 \times 10^3}{I_{xx}} \int_0^s t \cdot y \cdot ds + q_{s,0}$$

Taking moments about O to make $q_{s,0} = 0$.

$$\begin{aligned} I_{xx} &= \frac{3 \times 10^3 \times 8^3}{12} + (3 \times 10^3 \times 8) (0.433 \times 10^3)^2 \\ &+ \frac{2 \times 10^3 \times 8^3}{12} + (2 \times 10^3 \times \frac{12}{2}) (0.433 \times 10^3)^2 \\ &+ 2 \left[\frac{10 \times (10^3)^3 (\sin 60^\circ)^2}{12} \right] = 1025 \times 10^{10} \text{ or } 102.5 \times 10^8 \text{ mm}^4 \end{aligned}$$

Hence, equation 1 becomes,

$$q_s = - 4.87 \times 10^{-5} \int_0^s t \cdot y \cdot ds$$

In the wall O-A, $y = -0.43 \times 10^3$

$$q_{OA} = 4.87 \times 10^{-5} \int_0^s 12 \times 0.43 \times 10^3 \cdot ds = 0.25 S$$

Wall AB $y = -0.43 \times 10^3 s$

$$q_{AB} = -4.94 \times 10^{-5} \int_0^s 10(-0.43 \times 10^3 + 0.866 S_B) ds + q_{OA}$$



(29)

$$q_{AB} = -2.14 \times 10^{-4} S_B^2 + 250.$$

$$q_{OA} = 250 \text{ when } S = 1 \times 10^3$$

Wall BC $y = 0.4 \times 10^3 \times 10 \text{ mm}$

$$q_B = -4.94 \times 10^{-5} \int_0^s 8 \times 0.43 \times 10^3 ds + 246.$$

$$q_{AB} = 246 \text{ when } S_B = 1 \times 10^3 \text{ mm}$$

$$q_{BC} = -0.17 S_C + 246$$

When $S_C = 1.45 \times 10^3 \text{ mm}$.

$q_{BC} = 9 \text{ N/mm}$ which is almost getting to zero.

q_{BC} has to be zero, but the 9 N/mm is due to truncation error.